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**Detecting Model Drift in
LLM-as-a-Service APIs: A Benchmarking
Framework for Campaign-Relevance
Classification**

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Faculty of **Applied Computer Sciences and Biosciences**

BACHELOR THESIS

Detecting Model Drift in LLM-as-a-Service APIs: A Benchmarking Framework for Campaign-Relevance Classification

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Referat

This thesis addresses the challenge of maintaining stable performance in production-grade Large Language Model (LLM) classification pipelines. Based on two years of industrial application at Gocomo GmbH, the research focuses on *model drift*—the performance degradation caused by silent, third-party updates to API-based models.

The work contributes a three-part framework designed for high-volume marketing analytics. First, we provide a mathematical justification for reducing monitoring costs: by utilizing disagreement-based sampling, a high-sensitivity 300-sample monitoring subset is derived from a 1200-sample reference benchmark. This approach maximizes statistical power at a fraction of the original inference cost. Second, we formalize model selection as a multi-objective optimization problem, balancing predictive quality against operational expenditure. Finally, the thesis evaluates reflective prompt evolution (GEPA) as an adaptive mechanism to recover performance following detected drift events.

The resulting system, integrated with Langfuse and GitHub Actions, demonstrates that informative subsetting can act as a reliable leading indicator for model degradation. By combining statistical surveillance with automated adaptation, this framework provides a robust methodology for deploying non-stationary black-box models in business-critical environments.

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1 Introduction

1.1 Context and Motivation

Industrial classification pipelines increasingly rely on Large Language Model (LLM) providers, which operate as non-stationary black-box functions $f(x; \theta_t)$. Unlike static machine learning models, LLM-as-a-Service (LLMaaS) providers frequently update underlying parameters θ_t without notice. This temporal instability, or **model drift**, poses a significant risk to production systems: a model optimized at t_0 may exhibit regressive behavior at t_1 .

The practical motivation for this work stems from the author’s two-year tenure as a working student and internship at **Gocomo GmbH**, where these challenges were observed within the company’s core social-media relevance detection pipeline. In this environment, where high-volume data is processed in batches to determine the marketing relevance of social-media posts, manual verification is not scalable.

However, full-scale benchmarking on a large and statistically representative dataset is economically prohibitive for frequent monitoring. This creates a need for a statistically sensitive yet cost-effective surveillance framework.

1.2 Problem Statement

This thesis addresses the design of a monitoring system for black-box LLMs under non-stationary conditions. The core challenge is a dual-objective optimization:

1. **Statistical Efficiency:** Deriving a high-sensitivity monitoring subset that preserves the power to detect drift while minimizing sample size.
2. **Decision Utility:** Formulating a selection rule that balances the trade-off between classification accuracy and inference expenditure.

By integrating importance sampling with automated prompt adaptation (GEPA), this work provides a methodology for maintaining stable classification services in a provider-controlled API landscape.

1.3 Research Questions

This thesis answers the following research questions:

- RQ1:** How can the relevance detection task be formally represented as a binary classification problem in a non-stationary black-box environment?
- RQ2:** Can a disagreement-based subset $\mathcal{D}_n$ preserve the statistical sensitivity to model drift relative to a larger reference benchmark $\mathcal{D}_N$?
- RQ3:** What statistical criteria and hypothesis testing frameworks effectively detect significant performance degradation in a temporal loop?

- RQ4:** How should the model selection rule be formulated to optimize the trade-off between empirical risk and inference cost?
- RQ5:** To what extent can reflective prompt evolution (GEPA) serve as an adaptive recovery mechanism following a detected drift event?

2 Data and Evaluation Setup

2.1 Formal Problem Definition

To evaluate model drift rigorously, we must first define the classification task within a probabilistic framework.

2.1.1 The Probability Space

Let $\mathcal{X}$ denote the input space of social-media posts (textual content and associated metadata) and let $\mathcal{Y} = \{0, 1\}$ be the label space, where 1 indicates relevance to a campaign and 0 indicates irrelevance. We assume that pairs (x, y) are drawn from a joint distribution P_t on $\mathcal{X} \times \mathcal{Y}$. The subscript t signifies that this distribution or the underlying labeling function may evolve over time.

2.1.2 LLM as a Parameterized Conditional Distribution

In the context of LLM-as-a-Service, the model is a "black box." We define the classifier as a function f :

$$f(x; \pi, \theta_t) \rightarrow \hat{y} \in \{0, 1\} \quad (2.1)$$

where:

- $\pi \in \mathcal{P}$ represents the **prompt**, a discrete string chosen from the space of all possible instructions $\mathcal{P}$.
- θ_t represents the **frozen model weights** provided by the API at time t . We do not have access to θ_t , but we observe its outputs.

The model's output can be viewed as an approximation of the conditional probability $P(Y|X, \pi; \theta_t)$.

2.1.3 Loss Function and Risk

To quantify performance, we define a loss function $\mathcal{L} : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$. For this binary classification task, we utilize the zero-one loss:

$$\mathcal{L}(\hat{y}, y) = \mathbb{I}(\hat{y} \neq y) = \begin{cases} 1 & \text{if } \hat{y} \neq y \\ 0 & \text{if } \hat{y} = y \end{cases} \quad (2.2)$$

The **Expected Risk** (or generalization error) at time t for a fixed prompt π is:

$$R(f_{\pi, \theta_t}) = \mathbb{E}_{(x, y) \sim P_t} [\mathcal{L}(f(x; \pi, \theta_t), y)] \quad (2.3)$$

Since P_t is unknown, we estimate the risk using the **Empirical Risk** $\hat{R}$ over a finite dataset $\mathcal{D}$ of size n :

$$\hat{R}(f_{\pi, \theta_t}; \mathcal{D}) = \frac{1}{n} \sum_{i=1}^n \mathcal{L}(f(x_i; \pi, \theta_t), y_i) \quad (2.4)$$

Model drift is therefore defined as a statistically significant increase in $R(f_{\pi, \theta_t})$ over time t for a fixed prompt π .

2.2 The 1200-Example Reference Benchmark

We formalize the task as a mapping $g : \mathcal{X} \rightarrow \{0, 1\}$. The reference benchmark consists of $N = 1200$ manually labelled examples, establishing the ground truth for initial model selection. However, evaluating all candidate models against N examples weekly is economically prohibitive.

2.3 The 300-Example Monitoring Subset

To reduce operational costs, we construct a monitoring subset of $n = 300$ examples. Crucially, this is not a naive uniform random sample. It is an *informative* subset heavily weighted towards instances where models historically disagreed. Because it over-indexes on edge cases, it acts as a stress test. While it may not provide an unbiased estimate of global accuracy, it is highly sensitive to model drift.

2.4 Statistical Justification of the Monitoring Subset

The reduction from the reference benchmark ($N = 1200$) to the monitoring subset ($n = 300$) is justified by a shift from *population estimation* to *change detection*.

2.4.1 Informative Sampling via Model Disagreement

The monitoring subset was constructed by selecting instances where at least one model in the initial benchmark exhibited disagreement. Let y_i be the ground truth and $\hat{y}_{i,m}$ be the prediction of model $m \in \mathcal{M}$. The subset $\mathcal{D}_{300}$ is defined as:

$$\mathcal{D}_{300} \subset \{x_i \in \mathcal{D}_{1200} \mid \exists m_j, m_k \in \mathcal{M} : \hat{y}_{i,m_j} \neq \hat{y}_{i,m_k}\} \quad (2.5)$$

This approach is grounded in the principles of *Active Learning*, specifically *Uncertainty Sampling*. By focusing on the disagreement region, we sample from the portion of the manifold where the models' posterior probabilities $P(Y|X)$ are closest to the decision threshold.

2.4.2 Sensitivity and Statistical Power

From a hypothesis testing perspective, the goal of weekly monitoring is to reject the null hypothesis $H_0 : \Delta A = 0$ (no drift). The statistical power of a test—the probability of detecting drift when it exists—is higher on instances with high prediction variance.

In the full dataset ($N = 1200$), a large portion of instances are "easy" cases with $P(Y = 1|X) \approx 1$ or 0 . On these instances, the Bernoulli variance $p(1 - p)$ is minimal, making them "robust" to small model drifts. In contrast, the disagreement subset ($n = 300$) centers the evaluation on the "hard" cases where $p \approx 0.5$.

While $\hat{A}_{300}$ is a biased estimator of the global population accuracy, it serves as a **proxy metric** with higher sensitivity. A performance drop of 5% on the hard 300-sample subset is a leading indicator of degradation that would require a much larger (and more expensive) sample to detect on the full 1200-sample distribution. Thus, $n = 300$ provides an optimal balance between inference cost and the statistical power to detect model updates.

2.5 Theoretical Justification of Disagreement Sampling

The construction of the monitoring subset $\mathcal{D}_{300}$ via model disagreement represents a deliberate departure from simple random sampling (SRS). In this section, we formalise this approach through the lenses of *importance sampling* and *active learning*.

2.5.1 Selection Bias and the Sampling Indicator

Let $S_i \in \{0, 1\}$ be a Bernoulli random variable indicating whether the i -th instance from the reference benchmark $\mathcal{D}_N$ is included in the monitoring subset $\mathcal{D}_n$. Under SRS, $P(S_i = 1) = n/N$. In our framework, the inclusion probability is conditioned on the disagreement of a model ensemble $\mathcal{M}$:

$$P(S_i = 1 \mid x_i, \mathcal{M}) = q_i, \quad (2.6)$$

where q_i is high if the models in $\mathcal{M}$ yield conflicting predictions $\hat{y}_{i,m}$.

Consequently, the empirical accuracy on the subset, $\hat{A}_n$, is a **biased estimator** of the population accuracy A_N :

$$\mathbb{E}[\hat{A}_n] \neq A_N. \quad (2.7)$$

This bias arises because $\mathcal{D}_n$ intentionally over-represents "hard" cases near the decision boundary, for which the expected accuracy is lower than the population average.

2.5.2 Change of Measure and the Horvitz–Thompson Estimator

To reconcile the subset performance with the population mean, we can apply a **change of measure**. From the theory of *importance sampling*, an unbiased estimate of A_N can be recovered by weighting each observation by the inverse of its inclusion probability.

Let $q_i = P(S_i = 1 \mid x_i, \mathcal{M})$ denote the (possibly model-dependent) inclusion probability. The Horvitz–Thompson estimator of the population accuracy is

$$\hat{A}_{\text{HT}} = \frac{1}{N} \sum_{i \in \mathcal{D}_n} \frac{\mathbb{I}(\hat{y}_i = y_i)}{q_i}, \quad (2.8)$$

where the factor $1/q_i$ serves as an **importance weight**. In principle, this restores unbiasedness with respect to the original reference distribution. In practice, our production pipeline treats $\hat{A}_n$ primarily as a *monitoring signal*, but the existence of $\hat{A}_{\text{HT}}$ clarifies the statistical relationship between the curated subset and the underlying data-generating distribution.

2.5.3 Bias–Power Trade-off in Drift Detection

The motivation for accepting selection bias lies in a deliberate **bias–power trade-off**. For drift detection, the primary objective is not to obtain a perfectly calibrated estimate of global accuracy, but to maximise the *power* of a test for changes over time.

By concentrating on the high-entropy region around the decision boundary, we sample from those $x \in \mathcal{X}$ where the classifier is intrinsically uncertain and where small perturbations of the underlying model parameters θ_t are most likely to flip predictions. Intuitively, these are the points where the gradient of the loss with respect to θ_t is largest; small, otherwise imperceptible shifts in θ_t that would be masked by “easy” examples in $\mathcal{D}_N$ are amplified in $\mathcal{D}_n$.

Formally, we trade interpretability of $\hat{A}_n$ as a population accuracy for increased sensitivity of the drift signal:

1. **Bias Increase:** $\hat{A}_n$ systematically underestimates A_N because it is evaluated on a stress-test subset.
2. **Power Increase:** For a fixed evaluation budget n , the test statistic based on $\mathcal{D}_n$ has higher power to detect changes in accuracy than one based on an SRS of the same size, since it concentrates mass on inputs with high prediction variance.

In this sense, $\mathcal{D}_n$ is designed as an optimised *diagnostic instrument* for temporal instability, rather than as a representative sample of day-to-day traffic.

3 Drift Monitoring and Model Selection Methodology

3.1 Formalisation of Temporal Evaluation

Let $t \in \{1, \dots, T\}$ represent weekly evaluation intervals. For each model m , we track the empirical accuracy $\hat{A}_t$ on the monitoring subset $\mathcal{D}_{300}$. Because $\mathcal{D}_{300}$ consists of high-disagreement cases, $\hat{A}_t$ will be lower than the accuracy on the full benchmark, but it serves as our primary monitoring signal S_t .

3.2 Statistical Framework for Drift Detection

To distinguish between random performance fluctuation and genuine model drift, we utilize a threshold-based alerting rule.

3.2.1 Standard Error on the Informative Subset

Given that $n = 300$, the standard error of our accuracy estimate $\hat{A}_t$ is:

$$SE(\hat{A}_t) = \sqrt{\frac{\hat{A}_t(1 - \hat{A}_t)}{n}} \quad (3.1)$$

For a model with an expected accuracy of 0.70 on the hard subset, $SE \approx 0.026$ (2.6%).

3.2.2 The Alerting Rule

We define a performance drop as statistically significant if the difference between the current accuracy and the baseline accuracy $\hat{A}_0$ exceeds a confidence interval. However, for production stability, we use a **Practical Drift Threshold** τ_{drift} . An alert is triggered if:

$$\hat{A}_t < \hat{A}_0 - k \cdot SE(\hat{A}_0) - \tau_{\text{drift}} \quad (3.2)$$

where k is a scaling factor (typically $k = 2$ for a $\approx 95\%$ confidence interval) and τ_{drift} is the minimum degradation the business is willing to tolerate before intervening.

3.3 Accuracy–Cost Trade-off Modeling

Model selection is formulated as a multi-objective optimization problem. We define the **Operational Value Score** $V_t(m)$:

$$V_t(m) = \frac{\hat{A}_t(m)}{C(m)^\alpha} \quad (3.3)$$

where $C(m)$ is the cost per 1,000 requests and α is a sensitivity parameter.

- If $\alpha = 0$, we select purely based on accuracy.
- If $\alpha = 1$, we select based on "accuracy per dollar."

In our workflow, we set α based on the current budget constraints of the Gocomo pipeline.

3.4 Decision Logic for Model Switching

The system follows a three-step decision tree each week:

1. **Monitor:** Calculate V_t for all candidate models.
2. **Detect:** Check if the currently deployed model m_{curr} has triggered a drift alert.
3. **Act:** If drift is detected, calculate if another model m_{alt} provides a significantly higher V_t . If no model satisfies the threshold, trigger the **Prompt Optimisation** loop (Chapter 4).

3.5 Statistical Drift Detection

Let $\hat{A}_t$ be the empirical accuracy of a model on the monitoring subset at week t . We define the temporal performance delta as $D_t = \hat{A}_t - \hat{A}_{t-1}$. An alert is triggered if the degradation exceeds a practical operational threshold τ_{drift} :

$$D_t < -\tau_{\text{drift}} \quad (3.4)$$

This threshold must account for the natural variance of the 300-sample subset to avoid false-positive alerts.

3.6 Cost-Aware Model Selection

Predictive accuracy cannot be isolated from inference cost in a production environment. Let $C_t(m)$ be the inference cost and $A_t(m)$ be the accuracy for model m . We define a deployment utility function J_t that penalizes high costs:

$$J_t(m) = A_t(m) - \lambda C_t(m) \quad (3.5)$$

where $\lambda > 0$ represents the business's willingness to pay for marginal accuracy improvements. The system dynamically selects the model $m^* = \arg \max_m J_t(m)$ each week.

4 Prompt Optimisation as an Adaptive Mechanism

4.1 Integrating Optimisation into the Monitoring Loop

When a deployed model triggers a drift alert ($D_t < -\tau_{\text{drift}}$), switching to a more expensive model is not the only recourse. We can attempt to recover performance via prompt optimization.

4.2 Pareto-Based Prompt Search (GEPA)

We utilize a reflective evolutionary method (GEPA) for prompt adaptation. Because prompt space is discrete and non-convex, greedy selection often fails. Instead, we maintain a candidate pool of prompts and evaluate their per-instance score vectors $\mathbf{s}(\Pi) = (s_1(\Pi), \dots, s_n(\Pi))$.

We retain prompts based on Pareto dominance to preserve search diversity. A prompt $\Pi^{(a)}$ dominates $\Pi^{(b)}$ if it performs at least as well on all instances and strictly better on at least one. By mutating prompts on the Pareto frontier using natural-language reflections from execution traces, we can systematically search for a prompt Π^* that maximizes $J_t(m)$ without altering the underlying API model.

5 System Architecture and Experiments

5.1 Pipeline Implementation

The evaluation framework is orchestrated via GitHub Actions, running batch jobs against external LLM APIs. Prompts, predictions, and ground-truth labels are logged in Langfuse, allowing for full observability and dashboard integration.

5.2 Results: Monitoring, Costs, and Savings

[Placeholder for experimental results: Prove how much money is saved using the 300-subset vs 1200-subset, demonstrate a captured instance of model drift, and show the before/after accuracy of applying GEPA.]

6 Conclusion

This thesis provides a statistically grounded, cost-aware framework for the continuous evaluation of LLMs in production. By decoupling the initial reference benchmark from an informative monitoring subset, and pairing statistical drift detection with automated prompt optimization, organizations can maintain high classification performance despite the inherent instability of black-box LLM APIs.

Statutory Declaration in Lieu of an Oath

I – Georgii Burdin – do hereby declare in lieu of an oath that I have composed the presented work independently on my own and without any other resources than the ones given.

All thoughts taken directly or indirectly from external sources are correctly acknowledged.

This work has neither been previously submitted to another authority nor has it been published yet.

Mittweida, 12. March 2026

Location, Date

Georgii Burdin